# StockEvent Lens — Batch 006 Intelligence Dashboard

Articles 026–030 Analysis, Vector Embedding Sensitivity & Conflict Thresholds

Batch ID
`batch_06 (Articles 026-030)`

TEST 026
**Original** 100% Base
SpaceX Starmind AI satellites target 250 kW solar power with Nvidia NV72 racks[cite: 56, 60].

TEST 027
**Repeat** 96% Overlap
Re-syndicated fixture wire story detailing identical 250 kW solar array facts[cite: 57, 60].

TEST 028
**Update** +25% New Info
Secures regulatory approval, passes 300 kW peak load tests, target early Q1 2027 launch[cite: 58, 60].

TEST 029
**Conflict** Dispute (45 kW)
Contradicts power output: claims thermal flaws cap solar power at 45 kW per satellite[cite: 59, 60].

TEST 030
**Different** 0% Overlap
Unrelated announcement of Nvidia's $10B Healthcare AI initiative starting Group 012[cite: 55, 60].

## Technical Analysis: Dense Vector Blindness vs. Numerical Contradiction Metrics

A core challenge in automated financial news aggregation is that dense vector models (e.g., BAAI/bge-small-en-v1.5, OpenAI Ada-002) compute cosine similarity based on *semantic context* rather than *numerical fidelity*. Changing power capacity specifications (e.g., 250 kW vs 45 kW in test_029) yields high vector similarity (~0.88) because both texts share identical surrounding context tokens ("SpaceX", "Starmind", "solar power", "NVIDIA").

```
// Formula: Formal Conflict Detection Condition
Conflict_State = ( Cosine_Similarity(E_orig, E_cand) > τ_sim ) && ( ΔF_contradiction == 1.0 )
where τ_sim = 0.60, and ΔF_contradiction = 1.0 indicates a hard numerical or engineering spec dispute.
```

**1. Dense Embedding Blindness**
Dense embeddings project words into high-dimensional space where "250 kW" and "45 kW" share near-identical context embeddings. Thus, pure vector cosine distance fails to flag this metric conflict.

**2. LLM Evaluation Override**
To prevent missing critical engineering disputes, the LLM reasoning layer performs explicit entity-attribute-value matching, overriding high vector similarity when $\Delta F_{contradiction} = 1.0$ is detected.

## Article Relationship & Percentage Breakdown (Batch 006)

| FILE IDENTIFIER | PATTERN LABEL | KEYWORDS IDENTIFIED | SIMILARITY % | DELTA % | CLASSIFICATION RATIONALE |
|---|---|---|---|---|---|
| `test_026_original` | Original (%5=1) | SpaceX Starmind, 250 kW solar power, Nvidia NV72, Elon Musk, ISS comparison[cite: 56] | 100% | Baseline | Establishes breaking news frame on SpaceX Starmind AI satellites and Nvidia hardware integration[cite: 60]. |
| `test_027_repeat` | Repeat (%5=2) | Starmind AI fleet, 250 kW solar array, NV72 AI rack, space station benchmarks[cite: 57] | 96% | 0% New Facts | Rephrased wire report detailing identical power specs and hardware plans without new developments[cite: 60]. |
| `test_028_update` | Update (%5=3) | **Regulatory approval**, 300 kW peak load test, Q1 2027 orbital test flight[cite: 58] | 83% | +25% Operational Facts | Advances event state by securing regulatory approval, testing 300 kW peak load, and moving launch target to early Q1 2027[cite: 60]. |
| `test_029_conflict` | Conflict (%5=4) | **45 kW power cap dispute**, thermal dissipation flaws, upending Nvidia compute[cite: 59] | 88% Match | $\Delta F_{contradiction}$ = 1.0 | Directly disputes initial 250 kW solar power output, asserting thermal flaws limit power to just 45 kW[cite: 60]. |
| `test_030_different` | Different (%5=0) | Healthcare AI initiative, $10B supercomputer partnership, Blackwell architecture, hospital networks[cite: 55] | 0% | Group 012 Trigger | Zero domain overlap; triggers standalone Group 012 event cluster[cite: 60]. |

### Group 011 — Consolidated Event Summary

4 Articles (026–029)

SpaceX CEO Elon Musk outlined plans for the Starmind AI compute satellite constellation powered by NVIDIA NV72/VR72 AI racks ( Articles 026–027 )[cite: 60]. Program progress advanced as SpaceX secured regulatory approvals, completed 300 kW peak load power regulation tests, and accelerated orbital test flight targets to early Q1 2027 ( Article 028 )[cite: 60]. However, baseline engineering capacity remains heavily disputed: original reports cited solar power output exceeding 250 kW per spacecraft, whereas a conflicting disclosure claims severe thermal dissipation constraints cap power at just 45 kW , threatening the deployment of NVIDIA hardware ( Article 029 ; $\Delta F_{contradiction}$ = 1.0)[cite: 60].

✓ Update Preserved: 300 kW Test & Q1 2027 Schedule
✓ Conflict Preserved: 250 kW vs 45 kW Thermal Cap ✓ Duplicate Removed: Article 027 Consolidated

### Group 012 — Standalone Summary

1 Article (030)

NVIDIA CEO Jensen Huang announced a $10 billion partnership with a major healthcare consortium to deploy medical diagnostic supercomputers using Blackwell architecture across 500 hospital networks for real-time genomic sequencing ( Article 030 )[cite: 60].

✓ New Event Cluster Established ✓ Zero Entity Contamination from Group 011

## Batch 006 Raw Data Source Inspector

JSON Format Verification & Evidence Code Snippets

`test_026_original.json` — Group 011

```
{
  "id": 62126369,
  "headline": "Elon Musk Says SpaceX's Starmind AI Sat
  "symbols": ["NVDA", "SPCX", "T", "TMUS"],
  "summary": "Elon Musk says SpaceX’s Starmind AI sate
}
```

*Establishes Starmind & 250 kW baseline.*

`test_028_update.json` — Group 011

```
{
  "id": 990000027,
  "headline": "SpaceX Accelerates Starmind AI Satellit
  "symbols": ["NVDA", "SPCX", "T", "TMUS"],
  "summary": "SpaceX advances its Starmind AI satellit
}
```

*Adds 300 kW load test & Q1 2027 launch.*

`test_029_conflict.json` — Group 011

```
{
  "id": 990000028,
  "headline": "SpaceX Starmind Project Stalls as Engin
  "symbols": ["NVDA", "SPCX", "T", "TMUS"],
  "summary": "SpaceX's Starmind AI satellites face maj
}
```

*Disputes 250 kW vs 45 kW thermal cap.*